National Institute of Business Management

Programming Data Structures and Algorithms - 1
- Report -

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Introduction

This report and the project we have implemented represents a comprehensive technical analysis of the **Power Outage Tracking System**, a software solution developed to modernize the power grid breakdown response in Sri Lanka. By transitioning from a manual, subjective workflow to an automated system grounded in **Graph Theory, Hierarchical Sorting (Heaps), and Prefix-Based Retrieval (Tries)**, this tool achieves a significant reduction in response latency and operational overhead. The focus of this report is not on the user interface, but on the usage of fundamental data structures to solve real-world logistical inefficiencies.

1.1. Problem Definition: Existing CEB Outage Management System

The Ceylon Electricity Board (CEB) and other local utility providers currently manage outages through a process that is manual and fragmented:

1. **Subjective Prioritization**: Outage reports are often resolved on a first-come-first-served basis or through manual triage by operators. This ignores the "Criticality Index" (e.g., outages affecting hospitals or industrial zones vs. residential areas).
2. **The "Known Route" Fallacy**: Breakdown crews typically follow routes with which they are familiar. In a complex urban grid, the "known route" is rarely the mathematically shortest path. This leads to increased fuel consumption and delayed restoration.
3. **Search Inefficiency**: Finding consumer details in large databases using traditional search methods can be slow during peak outage periods (e.g., during monsoon storms).

1.2. Data Structure Implementations

To eliminate the inefficiencies in the existing system we have designed a new system by making the breakdown restoration process “algorithmic,” which utilizes carefully selected data structures and algorithms for each specified feature.

A. Tree (Trie – Prefix Tree) Data Structure: Phone Number Search

To manage the initial entry point of an outage report, we implemented a Trie (Prefix Tree).

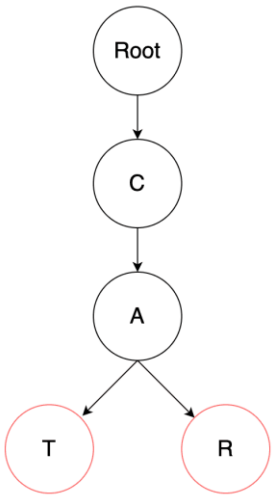
- #### Architectural Design

  Unlike a Hash Map, which requires a full key match, the Trie allows for incremental search (Search-as-you-type).

- **Node Structure**: Each **TrieNode** contains a dictionary of children and a **records** pointer.
- **Linked List Integration**: We utilized a **Linked List (RecordLink)** at each node to manage "collisions" or multiple records sharing the same phone number prefix.

- Algorithmic Logic

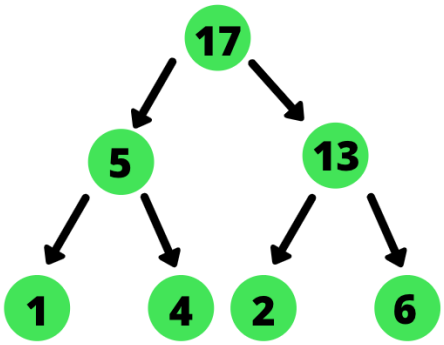
When the operator types "077", or any other first three letters from a phone number the algorithm traverses the tree to the '7' node and then performs a collect all operation.



- **Time Complexity:** Search is $O(m)$ where m is the length of the phone number, this guarantees m length searching operations regardless of how many millions of users are in the system.
- **Space Optimization**: The Trie shares common prefixes among phone numbers, reducing the overall memory footprint compared to storing every number as a full string in a flat array.

B. Max-Heap (Heap Sort): Outage Prioritization

In the existing system, prioritization is human-dependent. Our solution uses a Max-Heap to ensure the most critical task is always at the top of the queue.



- The Comparison Logic

  To create an accurate sorted list of outages, we have implemented the soring algorithm that uses multi-level comparison.

  - **Primary Key**: priority (1-10 scale).
  - **Secondary Key**: timestamp (Earlier reports get higher priority if levels are equal).

- Heap Sort Mechanism

  When the “Outage List” created, the system performs a **Heap Sort**. It builds a max heap from the active outages $O(N)$ and then repeatedly extracts the maximum $O(\log\ N)$, this guarantees $O(N\ log\ \ N)$ performance.

C. Graph Theory (Dijkstra's Algorithm): Shortest Path Finder

To find the shortest path from the central CEB hub (Node 65) to the outage site, we use **Dijkstra's Algorithm**.

- **Process**: The algorithm maintains a list of distances and previous nodes. It "relaxes" the edges of the graph, always picking the node with the minimum distance from the heap.
- **Efficiency**: This reduces the complexity from $O(V\text{^}2)$ to $O((E+V)\ log\ V)$. For a grid with 90+ nodes and hundreds of connections, the path is found in under 1ms.

1.3. Improvements and Enhancements over the Existing System

By comparing the old system with new enhanced system, below are the selling points that we have achieved.

- **Reduced Downtime**: By getting crews to the site faster and ensuring the most critical areas are fixed first, the overall System Average Interruption is lowered.
- **Fuel Efficiency**: Following the Dijkstra-calculated path reduces unnecessary mileage for heavy breakdown vehicles.
- **Auditability**: Every prioritization decision is based on a transparent algorithm (Heap Sort), removing accusations of mismanagement.
- **Data Integrity**: The use of structured DSA like Tries ensures that consumer data is retrieved accurately and quickly.

Conclusion

The Enhanced **Power Outage Tracking System** proves that optimized data structures are not just academic exercises but essential tools for infrastructure management. By modeling the power grid as a graph and the outage queue as a heap, we have created a system that is significantly more efficient than the manual methods currently prevalent in Sri Lanka. This tool provides a blueprint for how utility companies can leverage computer science to improve public service delivery.